# Melih Darcan

Computer Vision Researcher · Human Pose and Motion

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Google Scholar

## Summary

- MSc student in Artificial Intelligence Engineering at METU ImageLab, supervised by Emre Akbas. My research is on human pose and motion, and on making computer vision work for traditional folk dances: robust pose estimation, motion data, and the datasets that make both possible.

## Education

**MSc** **Middle East Technical University**, Artificial Intelligence Engineering — Ankara, Turkey
- ImageLab, Department of Computer Engineering. Supervised by Emre Akbas — Sept 2026 – present

**BSc** **Hacettepe University**, Artificial Intelligence Engineering — Ankara, Turkey
- GPA: 3.17 / 4.00 — Sept 2020 – Feb 2025
- Machine Learning, Deep Learning, Image Processing, Computer Vision, Bioinformatics

## Experience

**Inzva**, Guide & Participant — Istanbul, Turkey & Online
Aug 2023 – present · 3 years 4 months
- Observer in AI Projects #10, lightly guiding a team through the program
- Volunteer in Vision Cohort 2026, helping with logistics
- Proposed and led a pose estimation and dataset creation project for folk dances in AI Projects #9
- Guide for the Applied AI and Deep Learning study groups, lecturing on Video Understanding, Virtual Humans and CNNs
- Improved the DeTraffic project in AI Projects #8 with Q-Networks using Kolmogorov-Arnold Networks (KANs) for traffic control
- Completed the Google Developers ML Bootcamp (TensorFlow) and competed in Kaggle challenges

**Hacettepe University, Biological Data Science Laboratory**, Researcher — Ankara, Turkey
July 2022 – Oct 2025 · 3 years 4 months
- Developed, maintained and refactored pypath (Python, GraphQL, REST API), adding dataset modules that cut data preprocessing time for bioinformatics researchers
- Developed a GraphQL API for the CROssBAR knowledge graph database
- Worked on the LLM integration of CROssBAR
- Built the server infrastructure of CROssBAR

**Samsung Innovation Campus**, Guide & Participant — Ankara, Turkey & Online
Aug 2023 – Jan 2025 · 1 year 6 months
- Technical Guide for two consecutive terms (Jun. 2024, Jan. 2025), lecturing on model deployment strategies
- Co-developed DeTraffic, an AI-driven traffic management system aligned with UN Sustainable Development Goals 3, 11, 12 and 13
- Completed training on the AI development lifecycle, sustainable development and large-scale AI project methodology

## Publications

**HaNDF: Object-Conditioned Geometric Neural Hand Distance Fields** — 2026
Zuhao Liu, **Melih Darcan**, Zhengdi Yu, Riza Alp Guler, Tolga Birdal
circle-group.github.io/research/HaNDF (NeurIPS 2026)

**CROssBARv2: A unified computational framework for heterogeneous biomedical data representation and LLM-driven exploration** Apr 2026
Bünyamin Şen, Erva Ulusoy, **Melih Darcan**, Mert Ergün, Sebastian Lobentanzer, Ahmet S Rifaioglu, Dénes Türei, Julio Saez-Rodriguez, Tunca Doğan
10.64898/2026.04.12.718028 (bioRxiv preprint (oral presentation at ECCB 2025))

**OmniPath: integrated knowledgebase for multi-omics analysis** 2026
Dénes Türei, Jonathan Schaul, Nicolàs Palacio-Escat, Balázs Bohár, Yunfan Bai, Francesco Ceccarelli, Elif Çevrim, Macabe Daley, **Melih Darcan**, Daniel Dimitrov, et al.
(Nucleic Acids Research)

**Protein language models for predicting drug-target interactions: Novel approaches, emerging methods, and future directions** 2025
Atabey Ünlü, Erva Ulusoy, Melih Gökay Yiğit, **Melih Darcan**, Tunca Doğan
(Current Opinion in Structural Biology)

## Technical Skills

**Interests:** Virtual Humans, Computer Vision, Deep Learning, Machine Learning, Reinforcement Learning

**AI:** PyTorch, PyTorch3D, PyTorch Geometric, TensorFlow

**Programming:** Python, C, Java

**Other:** Linux system administration, Bash, Astro

**Languages:** Turkish (native), English (C1; IELTS 8/9, TOEFL 110/120)